

~~Handwritten text, mostly obscured by blue ink.~~

48/50

1) Write the noble gas ground-state electron configurations for the following atoms and ions

a) $[\text{Ne}] 3s^2 \checkmark$

b) $[\text{He}] 2s^2 2p^3 \checkmark$

c) $[\text{Ar}] 3s^2 3p^3 \checkmark$

d) $[\text{Xe}] 6s^2 4f^4 \checkmark$

e) $[\text{Ar}] 4s^2 3d^6 \checkmark$

2) Using Slater's rule, calculate the effective nuclear charge of the electrons in the following orbitals

a) $Z_{\text{eff}} = Z - \sigma$; $Z = 12$ Mg: $1s^2 2s^2 2p^6 3s$

$\sigma = 2 \cdot 1 + 8 \cdot 0,85 + 1 \cdot 0,35$

$\sigma = 9,15$

$Z_{\text{eff}} = 12 - 9,15$

$= \underline{2,85} \checkmark$

b) $Z_{\text{eff}} = Z - \sigma$; $Z = 7$ N: $1s^2 2s^2 2p^3$

$\sigma = 2 \cdot 0,85 + 4 \cdot 0,35$

$\sigma = 1,70 + 1,40$

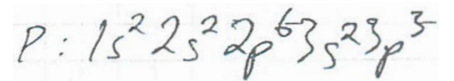
$\sigma = 3,1 \checkmark$

$Z_{\text{eff}} = 7 - 3,1$

$= \underline{3,9} \checkmark$

~~Grundlagen der Physik und Chemie~~
~~Klausur~~ ~~Vorbereitung~~

2) c) $Z_{\text{eff}} = Z - \sigma$; $Z = 15$



$\sigma = 2 \cdot 1 + 8 \cdot 0,85 + 4 \cdot 0,35$

$\sigma = 2,0 + 6,80 + 1,40$

$\sigma = 10,2$

$Z_{\text{eff}} = 15 - 10,2$
 $= 4,8$



d) Z_{eff}
 σ
 σ
 Z_{eff}

$= Z - \sigma$; $Z = 63$ Eu: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$

$= 2 \cdot 1 + 8 \cdot 1 + 18 \cdot 1 + 8 \cdot 1 + 10 \cdot 0,35$
 $= 40,55$

Ser lost für
 $(5s^2 5p^6 6s^2)$

$= 63 - 47,75 = 15,25$
 $= 15,25$

e) $Z_{\text{eff}} = Z - \sigma$; $Z = 28$ Ni²⁺: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8$

$\sigma = 18 \cdot 1 + 0,35 \cdot 7$

$\sigma = 18 + 2,45$

$\sigma = 20,45$

$Z_{\text{eff}} = 28 - 20,45 = 7,55$

Grundläggande uorganisk og organisk kemi

3) Answer the following questions and explain your reasoning, also by using your answers to the previous exercise.

h/ - Which is the largest atom between magnesium and phosphorus?

✓ The largest atom must be magnesium because of the electronegativity of phosphorus. Phosphorus is highly electronegative which in turn makes the surrounding electrons more compact and searching towards the nucleus than magnesium.

Slater's rule:

✓ - Which is the most electronegative between nitrogen and phosphorus?

That must be nitrogen because of how the periodic system is arranged; when one moves up and to the right the elements become more electronegative and vice versa when one moves in the opposite direction.

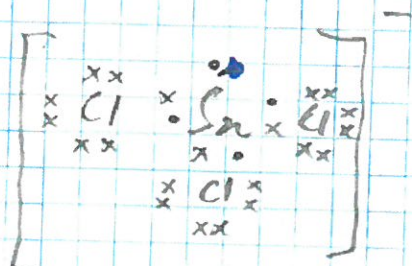
Grundlagen: unorganisch og organisk kemi

4) By taking in consideration the position of the elements in the periodic table, indicate the true sentences by a cross

- x Magnesium has a smaller atomic radius than sodium
- x The first ionization energy is higher for Mg than Na
- x The first ionization energy is higher for Na than Cs
- ~~x Mg has higher second ionization energy than Na~~
- x The attachment of one electron to a neutral Li atom is more favorable than to a neutral Be atom
- x In absolute values F has higher electron affinity than Li
- x Fluorine is more electronegative than iodine
- x $3+$ is the most common oxidation state for Al.

Grundlegende unorganisch og organisch kemi

5) Draw the Lewis structure of SnCl_3^-



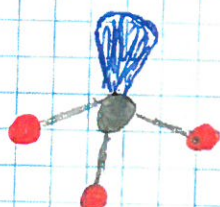
~~2~~ $\frac{4}{1}$

What is the bond order of this molecule?

- The bond order is 1,0 because of the covalent bonds between Sn and Cl, which are single bonds.

Determine the shape of this molecule by using the VSEPR model?

- It is a trigonal pyramidal because of the 3 bonding pairs and 1 lone pair.



Sketch

What is the hybridisation according to the Valence Bond model for this molecule?

- SnCl_3^- has a sp^3 hybridisation resulting in a tetrahedral molecular geometry.

Grundlagentheorie unorganisch. od. organisch Chem.

6) Draw the Lewis structure of NH_2^- .

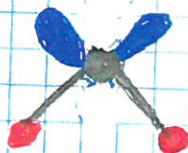


What is the bond order of this molecule?

- The bond order is 1,0 because of the single bonds.

Determine the shape of this molecule by using the VSEPR model?

- It has a V-shape because of the 2 bonding pairs and 2 lone pairs.



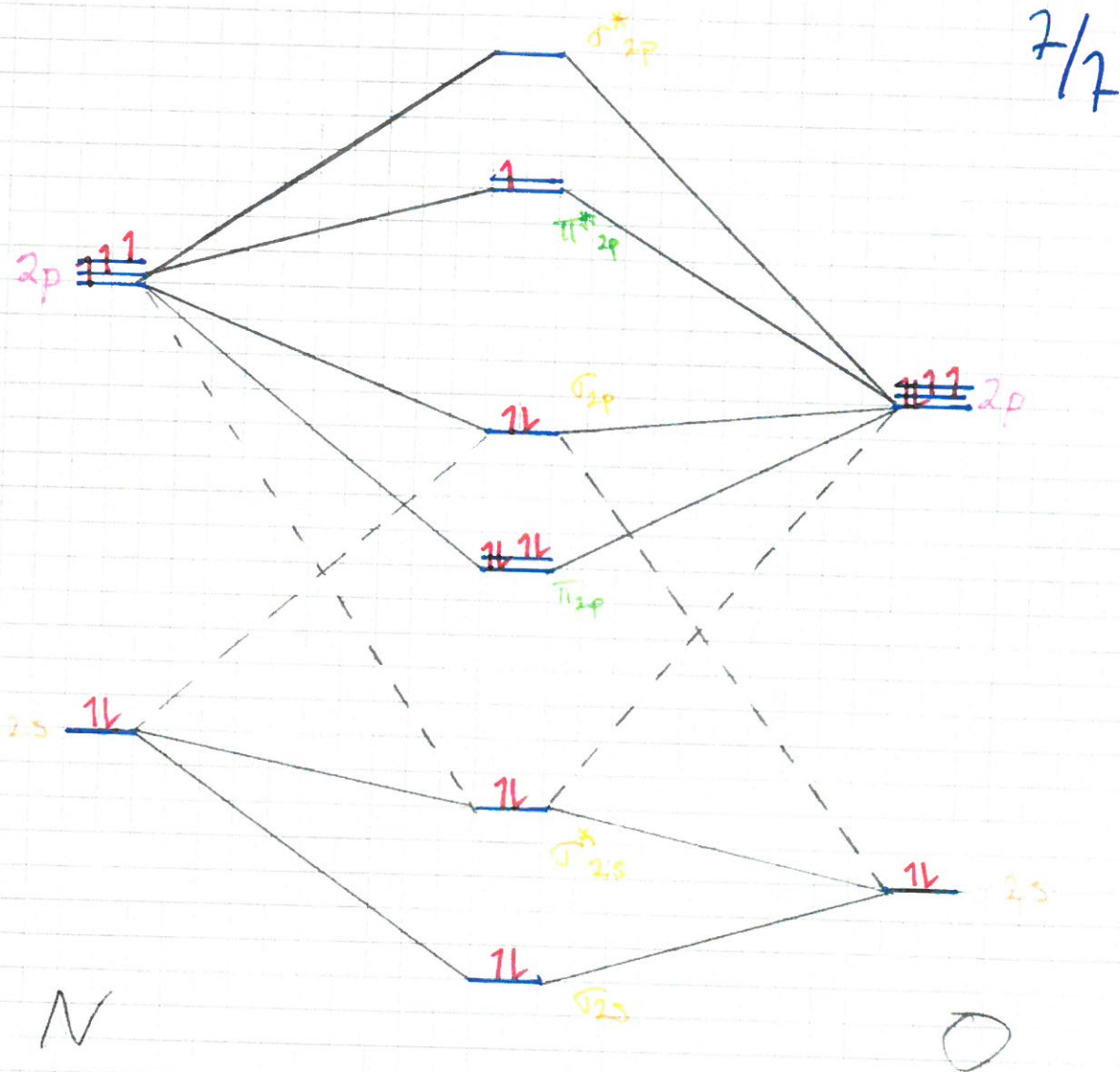
Sketch.

What is the hybridization according to the Valence Bond model for this molecule?

- It has a sp^3 hybridization resulting in a tetrahedral molecular geometry.

Grundläggande oorganisk og organisk kemi

- 7) Supposing that nitric oxide (NO) has a molecular orbital structure similar to carbon monoxide (CO) see below



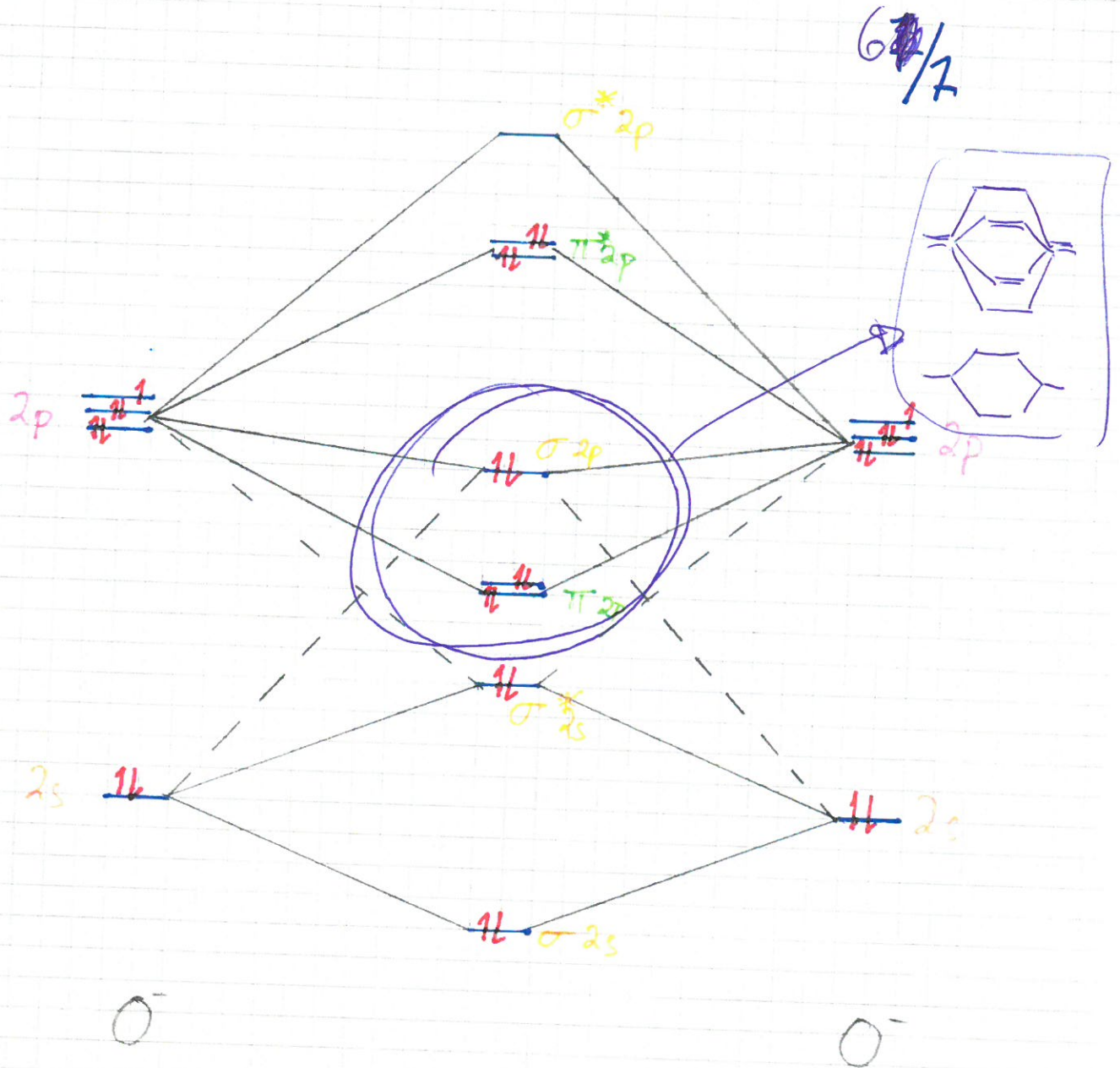
$$\text{bond order} : \frac{8 - 3}{2} = \frac{5}{2} = 2 \frac{1}{2}$$

→ σ - binding

→ π - binding

Grundlegende Vorgehensweise bei organischen Ionen

8) Use the molecular orbital theory to predict the bond order of O_2^{2-} ion.



σ → σ - binding
 π → π - binding